Burning Simple Polygons

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Abstract

Given a simple polygon P with n vertices and an integer k, we wish to find a set S of k vertices of P that minimize the maximum geodesic distance from any point in P to its geodesically closest vertex in S. We describe a dynamic programming algorithm that solves this problem in $O(n^7)$ time.

1 Introduction

We are given a simple polygon P defined by n vertices V, ordered clockwise around P. We are also given a positive integer $k \leq n$. The problem is to find a set U of k vertices in V that minimizes (over all size k subsets of V) the maximum geodesic distance (over all points $p \in P$) from U to p, where the distance from U to p is $d(U,p) \equiv \min_{u \in U} d(u,p)$ and $d(u,p)$ is the geodesic distance (length of the shortest path in P) from u to p. Alternatively, one can think of the set $U \subseteq V$ as being *ignition* points where we simultaneously ignite k fires that totally burn the flammable polygon in the least time.

Our problem is a restricted version (where $U \subseteq V$) of the general geodesic k-center problem in a simple polygon P where $U \subseteq P$. Oh et al. [7] described an $O(n^2 \log^2 n)$-time algorithm for the general problem in a simple polygon for $k = 2$. Soon after, Cho et al. [3] announced an optimal $O(n \log n)$ algorithm for this problem. The general problem for $k > 2$ seems more difficult. Evans and Lin [4] introduced the polygon burning problem and showed it to be NP-hard for polygons with holes. They also gave an algorithm that solves the problem in $O(kn^2)$ time for a restricted class of convex polygons. In this paper, we describe an algorithm to solve the problem for simple polygons in $O(n^7)$ time.

Our approach starts by identifying a small set of points in P, called the *potential final burn points*, denoted $\mathcal{F}$, that contains all points that could be the last to burn in P for any k and any $U \subseteq V$. The set $\mathcal{F}$ comprises precisely the vertices V, along with the points on the boundary of P equidistant from two vertices of P, and the points within P equidistant from three vertices of P. Thus $\mathcal{F}$ consists of points defined by one, two, or three vertices of P. Consequently, $|\mathcal{F}| \in O(n^3)$ and it can be calculated in $O(n^4)$ time [6]. For a point $p \in \mathcal{F}$ to be an actual final burn point of a set of $U \subseteq V$ of k ignition points, the vertex/vertices that define it must be in U and there must be a way to choose the remaining ignition points in U so that (1) U contains no vertex of P closer to p than p's distance, $t(p)$, to its defining points[1] (i.e. p is the last point to burn) and (2) every point in P is at distance at most $t(p)$ to its closest point in U (i.e. all of P burns). Given a particular value k, our algorithm considers each $p \in \mathcal{F}$, in increasing order of $t(p)$, as a candidate for the true final burn point for some set U of k ignition points. Using dynamic programming, our algorithm determines if such a set U exists. The first such p is the geodesic k-center.

2 Properties and Definitions

Notation: Let P be a simple polygon with vertices V. Let ∂P be the boundary of P and let $\partial P(a,b), a, b \in \partial P$ be the boundary of P clockwise from a up to b. For $u, v \in P$, let $\pi(u,v)$ be the shortest geodesic path in P from u to v and let $d(u,v)$ be the length of $\pi(u,v)$. The last vertex (or u if there is none) before v on $\pi(u,v)$ is called the *anchor* of v (with respect to u). Let $b(u,v)$ be the geodesic bisector of u and v in P. Given a set of ignition points (sites) $U \subseteq V$, let $\texttt{Vor}(U)$ be the geodesic nearest-point Voronoi diagram of U in P and let $\texttt{Vor}(U)[u]$ be the Voronoi region associated with $u \in U$ in $\texttt{Vor}(U)$. We also call $\texttt{Vor}(U)[u]$ the *burn region* associated with u.

Observation 1 *Since we are considering burn regions which are Voronoi regions of sites that are vertices of P, every burn region contains a vertex of P. In fact, burn regions are star-shaped[2] around their ignition point (from Aronov [1, Cor. 3.20]).*

Definition 1 (Burn Path) *A burn path $\pi(u,p)$ is the shortest geodesic path from an ignition point u to a point p in the burn region associated with u.*

Definition 2 (Final Burn Point) *A final burn point of a set of ignition points $U \subseteq V$ is a point $p \in P$*

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[1] A point p may have different values of $t(p)$ if it has different sets of defining points.

[2] A set $Q \subset P$ is *star-shaped* around $p \in P$ (with respect to the geodesic metric) if $\forall q \in Q : \pi(p,q) \subset Q$.

satisfying

$$d(U,p) = \sup_{q \in P} d(U,q)$$

For simplicity we will assume that P is in *general position*, meaning no vertex of P is equidistant from two other vertices of P, i.e., no bisector $b(u,v)$, $u,v \in V$ contains a vertex of P (from [1, Def. 3.21]). However, this assumption can be removed (See Appendix A). Note that P being in general position implies that all bisectors intersect the boundary of P at exactly two points.

Lemma 1 *Let $u,v \in V$. Then $d(u,p)$ where $p \in b(u,v)$ is strictly convex in p.*

Proof. See Appendix B for proof. □

Corollary 2 *Given the open subset S of $b(u,v)$ between two points p and q in $b(u,v)$*

$$\max\{d(u,p), d(u,q)\} > \sup_{p' \in S} d(u,p')$$

Proof. Since $d(u,p')$ is strictly convex in $p' \in b(u,v)$, the maximum value of $d(u,p')$ is attained at the boundaries of a given subdomain. □

Lemma 3 *Given a burn region R associated with ignition point u, there exists $p \in R$ such that $d(u,p) = \sup_{p' \in R} d(u,p')$. Moreover, for all such $p \in R$, at least one of the following is true:*

1. *$p \in (V \cap R) \setminus \{u\}$*
2. *p is the intersection of $b(u,v)$ with ∂P for some $v \in V$*
3. *p is the intersection of $b(u,v)$ with $b(u,w)$ for some $v,w \in V$*

Proof. See Appendix C for proof. □

Definition 3 (Potential Final Burn Point) *A potential final burn point is a point $p \in P$ such that:*

- *$p \in \bigcap_{n \in N} \texttt{Vor}(N)[n]$ for some $N \subseteq V$ where $1 \leq |N| \leq 3$.*
- *If $N = \{u\}$, $p \in V \setminus \{u\}$.*
- *If $N = \{u,v\}$, p is the intersection of $b(u,v)$ with ∂P.*
- *If $N = \{u,v,w\}$, p is the intersection of $b(u,v)$ with $b(u,w)$. (This condition, when $|N| = 3$, is redundant.)*

We use $\mathcal{F}$ to denote the set containing all such potential final burn points in P. For $p \in \mathcal{F}$, we call its corresponding N the set of defining ignition points of p.

Classifying points $p \in \mathcal{F}$ by the number of ignition points used to define them, we obtain a constructive definition of $\mathcal{F}$:

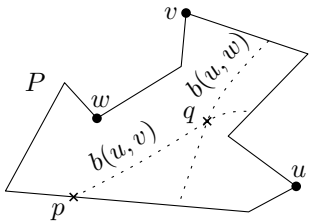


Figure 1: p is the Type II potential final burn point defined by u,v clockwise from u to v. q is the Type III potential final burn point defined by u,v,w.

A **Type I** potential final burn point is defined by a single ignition point ($|N| = 1$). All n vertices are Type I potential final burn points.

Type II: A Type II potential final burn point is defined by a pair of ignition points $N = \{u,v\}$. There are $\binom{n}{2}$ ways to choose distinct $u,v \in V$. For each pair, the bisector $b(u,v)$ intersects ∂P, yielding exactly two Type II potential final burn points (Fig. 1). Thus, we have up to $2\binom{n}{2}$ Type II potential final burn points.

Type III: A Type III potential final burn point is defined by a triple of ignition points $N = \{u,v,w\}$. There are $\binom{n}{3}$ ways to choose distinct $u,v,w \in V$. For each triple, if the corresponding Voronoi vertex exists in P, it constitutes a Type III potential final burn point (Fig. 1). (Note that potential final burn points can have multiple valid sets of defining ignition points.)

It is clear that $\mathcal{F}$ comprises precisely the vertices V, along with the points on the boundary of P equidistant from two vertices of P, and the points within P equidistant from three vertices of P.

Lemma 4 *$\mathcal{F}^* \subseteq \mathcal{F}$ where $\mathcal{F}^*$ is the set of final burn points of P for every possible set of ignition points $U \subseteq V$.*

Proof. Concretely,

$$\mathcal{F}^* = \bigcup_{U \subseteq V} \{p \in P | d(U,p) = \max_{q \in P} d(U,q)\}.$$

Fix $U \subseteq V$ and let $a \in \{p \in P | d(U,p) = \max_{q \in P} d(U,q)\}$. We must have $a \in \texttt{Vor}(U)[u]$ for some $u \in U$. Let R denote the burn region associated with u, $R = \texttt{Vor}(U)[u]$.

Let $b \in R$,

$$\begin{aligned}
d(u,b) &= d(U,b) && b \in \texttt{Vor}(U)[u] \\
&\leq \max_{q \in R} d(U,q) && b \in R \\
&\leq \max_{q \in P} d(U,q) && R \subseteq P \\
&= d(U,a) && \text{choice of } a \\
&= d(u,a) && a \in \texttt{Vor}(U)[u]
\end{aligned}$$

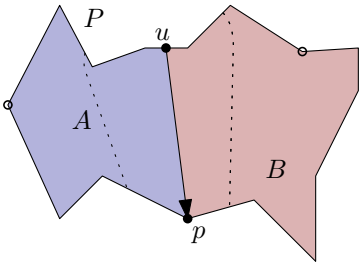


Figure 2: Enclosed regions $A = \mathcal{R}(\{u\}, p, (p, u))$ and $B = \mathcal{R}(\{u\}, p, (u, p))$ defined by ignition point u and a point p which lies on the boundary of P in $\texttt{Vor}(U)[u]$ $(U \subseteq V)$.

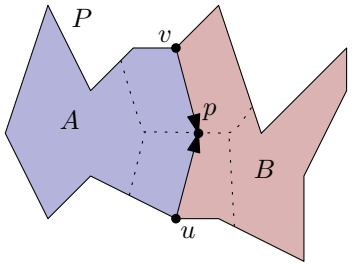


Figure 3: Enclosed regions $A = \mathcal{R}(\{u, v\}, p, (u, v))$ and $B = \mathcal{R}(\{u, v\}, p, (v, u))$ A and B defined by ignition points u, v and a point p which lies in the intersection of the burn regions associated with u and v in $\texttt{Vor}(U)$ $(U \subseteq V)$.

Therefore $d(u, a) = \sup_{b \in R} d(u, b)$ and a satisfies at least one of the three statements in Lemma 3.

If $a \in (V \cap R) \setminus \{u\}$, then we take $N = \{u\}$. If a is the intersection of $b(u, v)$ with ∂P for some $v \in V$, then we take $N = \{u, v\}$. If a is the intersection of $b(u, v)$ with $b(u, w)$ for some $v, w \in V$, then we take $N = \{u, v, w\}$. In all three cases it is clear that a satisfies Definition 3 and is therefore in $\mathcal{F}$.

Since each set in the union is a subset of $\mathcal{F}$, we have $\mathcal{F}^* \subseteq \mathcal{F}$ as desired. □

Note that for all final burn points $p \in \mathcal{F}^*$, the sets of defining points are covered by the sets of defining points for the same $p \in \mathcal{F}$ since we use every possible set.

2.1 Enclosed Regions

Given a set of ignition points $U \subseteq V$, an *enclosed region* is a region of P that is enclosed by a simple curve composed of one or two shortest paths from ignition points in U to some (shared) point p in their associated burn regions, and a continuous part of the boundary of P between the two end points of the curve. More specifically: Let $u \in U$ be an ignition point and $p \in \partial P$ be a point in $\texttt{Vor}(U)[u]$. The region enclosed by $\partial P(u, p)$ (or $\partial P(p, u)$) and $\pi(u, p)$ is an enclosed region with enclosing vertex u and enclosing point p (Fig. 2). Similarly, let $u, v \in U$ be ignition points such that $\texttt{Vor}(U)[u]$ and $\texttt{Vor}(U)[v]$ intersect at some point $p \in P$. The region enclosed by $\partial P(u, v)$ (or $\partial P(v, u)$) and the shortest paths $\pi(u, p)$ and $\pi(v, p)$ is an enclosed region with enclosing vertices u, v and enclosing point p (Fig. 3).

Let $\mathcal{R}(N, p, (a, b))$ denote the enclosed region of P enclosed by $\pi(N, p) \cup \partial P(a, b)$ where N is a set of one or two enclosing vertices, p is the enclosing point associated with the vertices, and $a, b \in \partial P$ are the end points of $\pi(N, p)$. Note that the ordered pair (a, b), with $a, b \in N \cup \{p\}$, determines which half of P is enclosed.

Lemma 5 (Isolation Property) *Given an enclosed region $R = \mathcal{R}(N, p, (a, b))$ where the paths $\pi(N, p)$ are burn paths in $\texttt{Vor}(U)$, no region in $\texttt{Vor}(U)$ exists on both sides of $\bigcup_{v \in N} \pi(v, p)$ (other than the regions associated with N).*

Proof. If the paths $\pi(N, p)$ are burn paths in $\texttt{Vor}(U)$, $\texttt{Vor}(U)$ contains regions associated with the ignition points in N. Furthermore, $\bigcup_{v \in N} \pi(v, p)$ is entirely contained in $\bigcup_{v \in N} \texttt{Vor}(U)[v]$ and must separate P into two halves. By Lemma 1, burn regions are simply connected. If there exists a region in $\texttt{Vor}(U)$ that is on both sides of $\bigcup_{v \in N} \pi(v, p)$ that isn't associated with vertices in N, it would imply it is not simply connected or that the region intersects other regions at more than just the boundary which contradicts our general position assumption. □

3 Algorithm Description

The input is a simple polygon P with vertices V, and the maximum number of vertex ignition points $k > 0$ allowed to burn P.

Definition 4 (Properly Burned) *Given a set of ignition points $U \subseteq V$, an enclosed region R is considered* properly burned *in time t if the burn paths enclosing R exist in $\texttt{Vor}(U)$ and R is entirely burned in time t.*

3.1 Preprocessing

We first calculate and store the set of all Type II and Type III potential final burn points (see Definition 3). Let $\texttt{F}_2[u, v]$ be the Type II potential final burn point $p \in \partial P(u, v)$, if it exists, with associated ignition points $u, v \in V$. Let $\texttt{F}_3[u, v, w]$[3] be the Type III potential final burn point p, if it exists, with associated ignition points $u, v, w \in V$. Populating these tables takes time $O(n^4)$ using the algorithm of Oh [6] which takes $O(n)$ time to compute the geodesic Voronoi diagram for a constant number of sites. It takes time $O(n^4 \log n)$ using the simpler algorithm of Asano and Asano [2].

[3]Assume different permutations of u, v, w index the same way.

3.2 Polygon Burning Algorithm

The Polygon Burning Algorithm returns the solution to the Polygon Burning Problem in P by finding a set of ignition points $U \subseteq V$, $|U| \leq k$, such that the time taken to burn the entirety of P is minimized. This is done by iterating through every point $p_0 \in \mathcal{F}$ in increasing order of associated final burn time, finding the smallest set U of ignition points that achieves that burn time, until we find U such that $|U| \leq k$.

(1) Let `DPTable` be a dynamic programming table, indexed by a time $t(p_0)$ and an enclosed region R, that stores the minimum size set of ignition points required to properly burn R in time $t(p_0)$.

(2) For each potential final burn point p_0 with associated set of ignition points $N = \{u\}$ (Type I), $N = \{u, v\}$ (Type II), or $N = \{u, v, w\}$ (Type III), in increasing order of $t(p_0) = d(u, p_0)$, we attempt to find the smallest set $U \subseteq V$ of ignition points that burn the entirety of P with final burn point p_0 and the burn paths from N to p_0 appear in $\texttt{Vor}(U)$. There are different initial enclosed regions for each type of potential final burn point:

Type I: There are two enclosed regions associated with a Type I potential final burn point: $\mathcal{R}(\{u\}, p_0, (u, p_0))$, $\mathcal{R}(\{u\}, p_0, (p_0, u))$. These correspond to the halves of the polygon clockwise of u up until p_0 and clockwise of p_0 up until u respectively which are separated by the burn path $\pi(u, p_0)$.

Type II: Without loss of generality, assume p_0 is clockwise of u up to v. There are three enclosed regions associated with a Type II potential final burn point: $\mathcal{R}(\{u\}, p_0, (u, p_0))$, $\mathcal{R}(\{v\}, p_0, (p_0, v))$, $\mathcal{R}(\{u, v\}, p_0, (v, u))$. These correspond to 3 regions of the polygon, one isolated by the burn path $\pi(u, p_0)$, one isolated by the burn path $\pi(v, p_0)$, and one isolated by the burn paths $\pi(u, p_0)$ and $\pi(v, p_0)$.

Type III: There are three enclosed regions associated with a Type III potential final burn point (note that we are assuming that u, v, w are in clockwise order around P): $\mathcal{R}(\{u, v\}, p_0, (u, v))$, $\mathcal{R}(\{v, w\}, p_0, (v, w))$, $\mathcal{R}(\{w, u\}, p_0, (w, u))$. These correspond to 3 regions of the polygon which are each enclosed by two burn paths from ignition points to p_0.

After determining the set of enclosed regions associated with p_0, we attempt to find the smallest set of ignition points as follows:

(a) We determine the sets $S_1, S_2, (S_3)$ corresponding to the optimal set of ignition points required to burn the enclosed regions associated with p_0 and N in time $t(p_0)$ using the ERA algorithm in Section 3.3.

(b) If $S_1 = \emptyset$ or $S_2 = \emptyset$ (or $S_3 = \emptyset$), this means it is not possible for p_0 to be a true final burn point so we continue to the next potential final burn point; otherwise

(c) let $U \leftarrow S_1 \cup S_2 (\cup S_3)$

Once we find U such that $|U| \leq k$, we return U.

Lemma 6 *If it is possible to burn P with a set of ignition points U such that p_0 is the final burn point and the paths from N to p_0 are burn paths in $\texttt{Vor}(U)$ then the algorithm returns the smallest such set. Otherwise it continues to the next potential final burn point.*

Proof. For each potential final burn point p_0 defined by a set of ignition points N, there exist known burn paths from N to p_0. Using these known paths, we define a set of enclosed regions S_R as in Step 2 such that each region in S_R is isolated from the rest by burn paths from N to p_0 and $\bigcup S_R = P$. Let $t(p_0) = d(N, p_0)$ (by definition, each ignition point in N is equidistant to p_0). Properly burning all of the regions in S_R in time $t(p_0)$ with a set of ignition points U is equivalent to P being entirely burnt in time $t(p_0)$, p_0 being a final burn point in P, and burn paths $\pi(N, p_0)$ being burn paths in $\texttt{Vor}(U)$. This is because properly burning all regions in S_R with U implies:

- The known burn paths from N to p_0 are burn paths in $\texttt{Vor}(U)$
- $\bigcup S_R = P$ is entirely burned
- Each region in S_R is burned in time less than or equal to $t(p_0)$ which means p_0, which is burned at exactly time $t(p_0)$, is the last point to burn.

Suppose U^* is a smallest set of ignition points that properly burns all of the regions in S_R. Since each region in S_R is isolated from the rest by known burn paths, U^* must properly burn each region in S_R optimally. By Theorem 9, in Step 2a, we obtain optimal sets for properly burning each region in S_R. Since U^* properly burns each region optimally, the union U of the optimal sets found in Step 2a has the same magnitude as U^* which implies U is optimal.

If it is not possible to burn P and maintain p_0 as the final burn point with burn paths from N to p_0, By Theorem 9, we will not be able to properly burn at least one of the regions in S_R which means we will continue to the next potential final burn point in Step 2b. □

Theorem 7 *This algorithm finds a set of ignition points U, $|U| \leq k$, such that for all points $p \in P$, the maximum geodesic distance from U to p is minimized.*

Proof. By Lemma 4, $\mathcal{F}^*$ is a subset of F_P. In addition, the set of defining points for each $p \in \mathcal{F}^*$ is covered

by the set of sets of defining points for corresponding $p' \in \mathcal{F}$. Suppose U^*, $|U^*| \leq k$, is an optimal set of ignition points that minimizes the maximum distance from an ignition point to a point $p \in P$. U^* will have an associated final burn point $p_0 \in P$. Since $\mathcal{F}$ covers every possible set of initial ignition points that make p the final point to burn in F (Lemma 4), if we look at potential final burn point p_0 with initial ignition points N such that the burn paths from N to p_0 exist as burn paths in $\texttt{Vor}(U^*)$, we will return a set U such that $|U| = |U^*|$ by Lemma 6.

Since U^* is optimal and we explore potential final burn points in increasing order of burn time, if we return before we get to potential final burn point p_0 with initial ignition points N, we must have returned a set U, $|U| \leq k$, with burn time equal to the burn time of U^*. □

3.3 Enclosed Region Algorithm (ERA)

The purpose of this algorithm is to fill in the dynamic programming table by finding the set with the minimum number of ignition points required to properly burn an enclosed region R of P in time $t(p_0)$. If it is not possible to properly burn R in time $t(p_0)$, the table is assigned the empty set.

Each subproblem is defined by an enclosed region $R = \mathcal{R}(N = \{u, (v)\}, p, (a, b))$ and a time $t(p_0)$ to burn R. Note that we assume $d(N, p) \leq t(p_0)$.

(1) Return $\texttt{DPTable}[t(p_0), R]$ if it has already been calculated; otherwise

(2) Determine if R is entirely burned in time $t(p_0)$ by the set N of enclosing vertices in linear time (see Section E). If R is entirely burned, we update the dynamic programming table: $\texttt{DPTable}[t(p_0), R] \leftarrow N$ and return N; otherwise

(3) Let $S_{best} \leftarrow \emptyset$ be the minimum set of ignition points required to burn R.

(4) If $|N| = 2$, attempt to recursively find the optimal sets S_A and S_B for enclosed regions A and B implied by Case 2 of Lemma 12. If S_A and S_B are both nonempty, $S_{best} \leftarrow S_A \cup S_B$.

(5) For each vertex $v' \in V \cap \partial P(a, b)$, we attempt to add v' as the next enclosing vertex.

  (a) Determine the enclosing point p' and enclosing regions A and B associated with adding v' as the next enclosing vertex (see Section 3.4). If we cannot add v' as the next enclosing vertex, we continue to the next iteration of the loop; otherwise

  (b) Recursively find optimal ignition point sets S_A and S_B that properly burn A and B respectively in time $t(p_0)$.

  (c) If $S_A = \emptyset$ or $S_B = \emptyset$, A and/or B could not be properly burned in time $t(p_0)$ so we continue to the next iteration; otherwise

  (d) If $|S_A \cup S_B| < |S_{best}|$ or $S_{best} = \emptyset$, $S_{best} \leftarrow S_A \cup S_B$.

(6) Update the dynamic programming table $\texttt{DPTable}[t(p_0), R] \leftarrow S_{best}$ and return S_{best}.

Lemma 8 *In ERA Step 4, given that burn paths $\pi(u, p)$ and $\pi(v, p)$ exist, properly burning A and B is equivalent to properly burning R, subject to $b(u, v)$ not intersecting any region in R other than $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$.*

Proof. If A and B are properly burned, burn paths $\pi(u, p')$ and $\pi(v, p')$ exist and are burned in time $t(p_0)$. This means burn paths $\pi(u, p)$ and $\pi(v, p)$ are burn paths in $\texttt{Vor}(U)$ since they are separated from $A \cup B$ by $\pi(u, p') \cup \pi(v, p')$. Furthermore, $R_{u,v} = R \backslash (A \cup B)$ which is enclosed by $\pi(u, p')$, $\pi(u, p)$, $\pi(v, p')$, and $\pi(v, p)$ is entirely burned by u and v (Obs. 5). This means $R = A \cup B \cup R_{u,v}$ is entirely burned in time $t(p_0)$ which implies R is properly burned. □

Theorem 9 *ERA finds a minimum set of ignition points to properly burn $R = \mathcal{R}(N = \{u, (v)\}, p, (a, b))$ in time $t(p_0)$, if it is possible. Otherwise, ERA returns the empty set.*

Proof. Suppose there exists some optimal set $U \subseteq V$ of ignition points that properly burn R in time t.

- If R is entirely burned by N in time $t(p_0)$, N is the optimal set since we don't need to add more ignition points and the enclosing vertices and enclosing point of R are preserved by default (Step 2).
- If R is not entirely burned by N in time t, there are three cases:

Case $N = \{u\}$: Without loss of generality, suppose $R = \mathcal{R}(\{u\}, p, (p, u))$. By Lemma 11, there exists an ignition point $v' \in U \cap \partial P(p, u)$ and a point $p' \in b(u, v') \cap \partial P(p, v')$ such that $\partial P(p, p')$ is entirely in $\texttt{Vor}(U)[u]$. Since $U \cap \partial P(a, b) \subseteq V \cap \partial P(a, b)$, at some point, we will attempt to add v' as the next enclosing vertex (ERA step 5). Since the enclosed regions A and B implied by adding v' exist in the context of U, by Theorem 10, we will obtain A and B such that properly burning A and B is equivalent to R being properly burned with v' as the next enclosing vertex (ERA step 5a). We recursively find optimal ignition sets S_A and S_B that properly burn regions A and B. Since A and B are isolated from each other in U by $\pi(v', p')$, they must be burned optimally in U or there would exist a set of ignition points more optimal than U. This implies $|S_A \cup S_B| = |U|$. Since U is optimal, the set $S_A \cup S_B$ of ignition points returned by the algorithm is optimal.

Case $N = \{u, v\}$ and $b(u, v)$ intersects regions other than $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$ in R: Without loss of generality, suppose $R = \mathcal{R}(\{u, v\}, p, (v, u))$. By Lemma 12 Case 1, there exists $v' \in U \cap \partial P(v, u)$ and $p' \in R$ such that p' is the Voronoi vertex of u, v, v' (it is equidistant from all three vertices). In a similar manner to the previous case, we will eventually attempt to add v' as the next enclosing vertex and find the regions A and B implied by adding v' as the next enclosing vertex (Obs. 4). We recursively find the optimal sets S_A and S_B that properly burn A and B. Since A and B are isolated from each other by $\pi(v', p')$, they must be burned optimally in U which implies $|S_A \cup S_B| = |U|$. Since U is optimal, the set $S_A \cup S_B$ of ignition points returned by the algorithm is optimal.

Case $N = \{u, v\}$ and $b(u, v)$ does not intersect any region in R other than $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$: In Step 4, we find the regions A and B implied by this case as shown in Case 2 of Obs. 4. Lemma 8 says that properly burning A and B is equivalent to properly burning R in this case. Since it must be possible to properly burn A and B (since U exists), we recursively find the optimal sets S_A and S_B that properly burn A and B. Since A and B are isolated from each other by $b(u, v)$, they must be burned optimally in U which implies $|S_A \cup S_B| = |U|$. Since U is optimal, the set $S_A \cup S_B$ of ignition points returned by the algorithm is optimal.

If there exists no way to properly burn R with some set $U \subseteq V$, for each v', either there will be no way to find A and B (Theorem 10 for Case 1,2, Lemma 8 for Case 3) or we will not be able to recursively find sets that properly burn A and B which means S_{best} will never be updated. Since S_{best} is never updated, we will fill the table with the empty set. □

3.4 Next Enclosing Vertex Algorithm

Given an enclosed region $R = \mathcal{R}(N, p, (a, b))$, a maximum time $t(p_0)$ allowed to burn R, and a potential next enclosing vertex $v' \in \partial V(a, b)$, we determine if v' can be added as the next enclosing vertex and we find enclosed regions A and B associated with adding v'. We assume that the burn paths enclosing R are already burned in time $t(p_0)$.

(1) If $d(v', p) < d(N, p)$, return $\emptyset$; otherwise

(2) There are two cases:

Case 1: $N = \{u\}$: Without loss of generality, assume R is bounded by $\partial P(p, u)$. We assume v' is the vertex described in Lemma 11 which means $p' = \texttt{F}_2[u, v']$ since p' is the intersection of $b(u, v')$ with ∂P on the same side of ∂P as p.

(a) If $d(v', p') > t(p_0)$, return $\emptyset$; otherwise

(b) If $\partial P(p, p')$ is not covered by u in time $t(p_0)$ (see Section E), return $\emptyset$; otherwise

(c) return A and B as in Obs. 3.

Case 2: $N = \{u, v\}$: Without loss of generality, assume R is bounded by $\partial P(v, u)$. We assume v' is the vertex described in Case 1 of Lemma 12 which means $p' = \texttt{F}_3[u, v, v']$ since it is equidistant from all three vertices (if p' does not exist, return $\emptyset$).

(a) If $d(v', p') > t(p_0)$, return $\emptyset$; otherwise

(b) return A and B as in Obs. 4.

Theorem 10 *Given an enclosed region $R = \mathcal{R}(N, p, (a, b))$, a time $t(p_0)$ and v' as the next enclosing vertex, if it is possible, the Next Enclosing Vertex Algorithm gives us two enclosed regions A and B separated by $\pi(v', p')$, such that A and B being properly burned in time $t(p_0)$ by a set of ignition points U is equivalent to R being properly burned in time $t(p_0)$ and paths $\pi(N, p')$ and $\pi(v', p')$ being burn paths in $\texttt{Vor}(U)$. If it is not possible the Next Enclosing Vertex Algorithm returns nothing.*

Proof. See Section D. □

3.5 Runtime Analysis

There are $O(n^3)$ possible values of t for indexing into the dynamic programming table. At each step of the recursion, we choose an enclosed region such that the enclosing point is a potential final burn point of P. This is true for the Polygon Burning Algorithm and in each case of the Next Enclosing Vertex Algorithm. So there are $O(n^3)$ possible enclosed regions meaning the total size of the dynamic programming table is $O(n^6)$. Each value in the table is filled in by one call to ERA. Each iteration of the loop in ERA takes constant time if we precalculate region coverage for vertices ($O(n^6)$ with brute force) and the set of all pairwise distances between $v \in V$ and $\mathcal{F}$ ($O(n^4 \log n)$ using [5]). Since ERA performs $O(n)$ iterations, the final run time of the algorithm is $O(n^7)$.

4 Open Problems

It seems likely that the running time of this algorithm can be improved. It is also likely that a slightly more complicated algorithm could handle a constant number of ignition points in the interior of polygon P.

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Appendix

A Removing the General Position Assumption

Assuming no vertex of P is equidistant to two other vertices ensures that the (geodesic) bisector of any two vertices of P is a 1D curve that intersects ∂P at exactly two points. Without this general position assumption, we can get positive area intersections of Voronoi cells as shown in Figure 4. To handle this, we define the bisector $b(u, v)$ to be only the 1D curve portion of the intersection between Voronoi cells of u and v in $\texttt{Vor}(U)$ (including the two intersection points on ∂P at the ends of the curve).

Observation 2 *All points in P on the same side of $b(u, v)$ as u are at least as close to u as they are to v*

Proof. If there exists a point p such that $d(u, p) > d(v, p)$ and $\pi(v, p)$ crosses $b(u, v)$ at a point x where $d(u, x) = d(v, x)$ then $d(u, p) \leq d(u, x) + d(x.p) = d(v, x) + d(x, p) = d(v, p)$ is a contradiction. □

Assume $b(u, v)$ intersects ∂P at some vertex w. If this bisector appears when running ERA, it must be the case that we are attempting to include both u and v in our solution which means we can assume any non-trivial intersection region is covered by the ignition point on the same side of $b(u, v)$ as the intersection region by Obs. 2.

B Proof of Lemma 1

Lemma 1 *Let $u, v \in V$. Then $d(u, p)$ where $p \in b(u, v)$ is strictly convex in p.*

Proof. The tangent to $b(u, v)$ bisects the angle between the directions to u and v of the shortest paths $\pi(p, u)$ and $\pi(p, v)$[4]. Let u' and v' be the anchor vertices along $\pi(p, u)$ and $\pi(p, v)$ respectively. Let the angle θ be the angle between $\overline{pu'}$ and $\overline{pv'}$ clockwise from $\overline{pv'}$ to $\overline{pu'}$.

[4] from [1, Lemma 3.22]

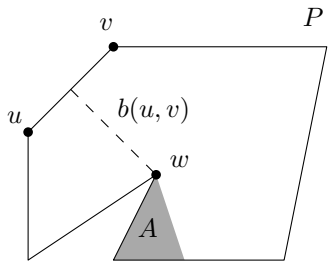


Figure 4: Polygon P is not in general position since $b(u, v)$ intersects vertex w of P. In this case, all points in region A are equidistant from both u and v.

Let $\alpha = \theta/2$ be the angle between the tangent to the bisector at p and $\overline{pu'}$. $\alpha \in [0, \pi]$ since, if this were not the case, then the shortest path from u to p would properly intersect $b(u, v)$. Consider point $p' \in b(u, v)$ infinitesimally close to p in the direction such that p' is clockwise from $\overline{pv'}$ to $\overline{pu'}$. We consider the segment of $b(u, v)$ between p and p' to be a straight line. Let β be the angle between the tangent of $b(u, v)$ at p' and $\overline{p'u'}$. We now have a triangle composed of segments and angles $\overline{pu'}, \alpha, \overline{pp'}, \beta, \overline{p'u'}, \varepsilon$ as shown in Fig. 5. This implies:

$$d(u', p) = \frac{\sin\alpha}{\sin\beta} d(u', p') \qquad \text{Law of Sines}$$

$$\beta = \pi - \alpha - \varepsilon \qquad \text{sum of angles}$$

$$\implies d(u', p) = \frac{\sin\alpha}{\sin(\alpha + \varepsilon)} d(u', p')$$

This implies $d(u', p) > d(u', p')$ for $\alpha < \pi/2$ and $d(u', p) < d(u', p')$ for $\alpha \geq \pi/2$ which implies the same for $d(u, p)$ and $d(u, p')$. Since alpha strictly increases as we move along $b(u, p)$ in the direction of p' from p (increases by ε at each step), $d(u, p)$ is strictly decreasing for $\alpha < \pi/2$ and strictly increasing for $\alpha \geq \pi/2$. □

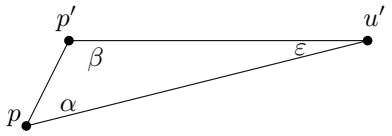


Figure 5: The triangle created by u', p and p' as in Lemma 1.

C Proof of Lemma 3

Lemma 3 *Given a burn region R associated with ignition point u, there exists $p \in R$ such that $d(u, p) = \sup_{p' \in R} d(u, p')$. Moreover, for all such $p \in R$, at least one of the following is true:*

1. *$p \in (V \cap R) \setminus \{u\}$*
2. *p is the intersection of $b(u, v)$ with ∂P for some $v \in V$*

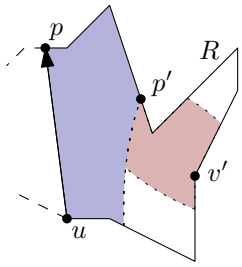


Figure 6: An enclosed region $R = \mathcal{R}(\{u\}, p, (p, u))$ as defined in Lemma 11

3. *p is the intersection of $b(u, v)$ with $b(u, w)$ for some $v, w \in V$*

Proof. Fix u and its corresponding burn region R. Existence of p follows directly from 1) the function $d(u, p) : R \longrightarrow \mathbb{R}$ is continuous and 2) R is closed and bounded in $\mathbb{R}^2$ and hence compact. Let $p \in R$ and suppose none of the above is true.

Case 1: p is an interior point of R. In this case, a point $q \in R$ with $d(u, q) > d(u, p)$ can be found by extending the last segment in $\pi(u, p)$ by an infinitesimally small amount and taking the new endpoint as q.

Case 2: p is not an interior point of R. Then $p \in b(u, v)$ for some $v \in V$ or $p \in \partial P$. If p is on some bisector then Lemma 2 implies the existence of $q \in R$ infinitesimally close to p on the same bisector as p such that $d(u, q) > d(u, p)$.

Suppose $p \in \partial P$ and the anchor vertex p' doesn't change at $\pi(u, p)$. We must have $|p'q| > |p'p|$ for all q lying on at least one side of p on the same polygon edge. Then we can find $q \in R$ on the polygon edge infinitesimally close to p such that $|p'q| > |p'p|$ while maintaining p' as the anchor vertex along $\pi(u, q)$. Then $d(u, q) = d(u, p') + |p'q| > d(u, p') + |p'p| = d(u, p)$ and $d(u, p) \neq \sup_{p' \in R} d(u, p')$.

Alternatively, if $p \in \partial P$ and the anchor vertex does change at $\pi(u, p)$. Let p' denote the anchor vertex and p'' denote the second to last vertex on $\pi(u, p)$. Similarly, we must have $|p''q| > |p''p|$ for all q lying on at least one side of p on the same polygon edge. We can find $q \in R$ infinitesimally close to p such that $|p''q| > |p''p|$ and the anchor vertex along $\pi(u, q)$ is either p' or p''. If the anchor vertex is p'' then by argument similar to the previous case, we are done. Otherwise, applying the triangle inequality under Euclidean distance, we have $d(u, q) = d(u, p'') + |p''p'| + |p'q| \geq d(u, p'') + |p''q| > d(u, p'') + |p''p| = d(u, p)$.

$d(u, p) \neq \sup_{p' \in R} d(u, p')$ in all cases if p satisfies none of the above statements. Combined with the fact that some $p \in R$ must satisfy $d(u, p) = \sup_{p' \in R} d(u, p')$, the lemma is true. □

D Algorithm Correctness

Lemma 11 *(see Fig. 6) Given an enclosed region $R = \mathcal{R}(\{u\}, p, (p, u))$, if R does not lie entirely in $\texttt{Vor}(U)[u]$, there exists an ignition point $v' \in \partial P(p, u)$ such that the following properties hold:*

- $\texttt{Vor}(U)[v'] \subset R$
- $p' \in \partial P(p, v') \cap b(u, v')$
- $\partial P(p, p') \subset \texttt{Vor}(U)[u]$

The symmetric claim holds for $R = \mathcal{R}(\{u\}, p, (u, p))$.

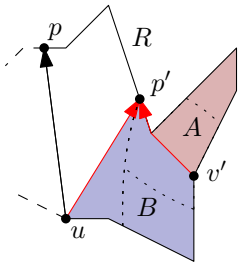


Figure 7: The enclosed regions implied by the existence of v' and p' in Obs. 3. $A = \mathcal{R}(\{v'\}, p', (p', v'))$, $B = \mathcal{R}(\{u, v'\}, p', (v', u))$.

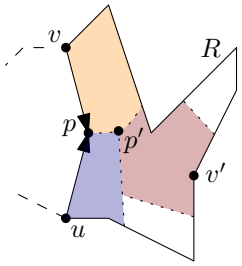


Figure 8: An enclosed region $R = \mathcal{R}(\{u, v\}, p, (v, u))$ that falls into Case 1 of Lemma 12.

Proof. Since R does not lie entirely in $\texttt{Vor}(U)[u]$, there exists a set S of one or more burn regions in R associated with ignition points in $U \setminus \{u\}$. Furthermore, by Lemma 5, since $\pi(u, p)$ separates P, the ignition points must all exist in $\partial P(p, u)$. By Obs. 1, there exist points along $\partial P(p, u)$ that are in $\bigcup S$. Since $p \in \texttt{Vor}(U)[u]$, as we travel along $\partial P(p, u)$ from u to p, we must eventually get to a point p' where $p' \in \bigcup S$ and, for all points $b \in \partial P(p, p')$, $b \in \texttt{Vor}(U)[u]$. We choose v' to be the ignition point associated with a region in S that contains p'. Because $\texttt{Vor}(U)[v']$ does not contain u, it is a proper subset of R. □

Observation 3 *(see Fig. 7) Given the enclosed region $R = \mathcal{R}(\{u\}, p, (p, u))$, the existence of p' and paths $\pi(u, p')$ and $\pi(v', p')$ in R implies the existence of two new enclosed regions derived from $\texttt{Vor}(U)$:*

- $A = \mathcal{R}(\{v'\}, p', (p', v'))$
- $B = \mathcal{R}(\{u, v'\}, p', (v', u))$

The symmetric claim holds for $R = \mathcal{R}(\{u\}, p, (u, p))$.

Lemma 12 *Given an enclosed region $R = \mathcal{R}(\{u, v\}, p, (a, b))$, if R does not lie entirely in $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$, there are two possible cases:*

- *Case 1 (Fig. 8): If $b(u, v)$ intersects some region other than $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$ in R, there exists an ignition point $v' \in U$ such that the following properties hold:*

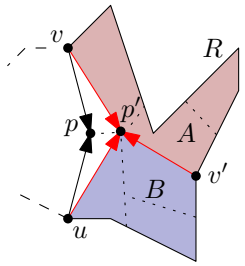


Figure 9: The enclosed regions implied by the existence of v' and p' in Case 1 of Obs. 4. $A = \mathcal{R}(\{v, v'\}, p', (v, v'))$, $B = \mathcal{R}(\{u, v'\}, p', (v', u))$.

  - $\texttt{Vor}(U)[v'] \subset R$
  - $\texttt{Vor}(U)[v']$ *intersects* $\texttt{Vor}(U)[u] \cap \texttt{Vor}(U)[v]$ *at some point* $p' \in R$
- *Case 2:* $b(u, v)$ *does not intersect any region in* R *other than* $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$. *The following properties hold:*
  - *There exists burn paths* $\pi(u, p')$ *and* $\pi(v, p')$ *where* p' *is the point at the intersection of* $b(u, v)$ *with* ∂P *in* R

Proof. Case 1: If $b(u, v)$ intersects regions other than $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$, there exists a region $\texttt{Vor}(U)[v']$ at the point p' on $b(u, v)$ at the boundary of $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$. This point, by definition, is the Voronoi vertex associated with vertices u, v, v' and therefore is in all three associated regions. Since $\pi((, u), p) \cup \pi(v, p)$ separates R from the rest of P, $\texttt{Vor}(U)[v'] \subset R$.

Case 2: $b(u, v)$ intersects $\partial P \cap R$ at some point p' since, if this were not the case, we could draw a path from u to v through R that does not intersect $b(u, v)$. Since $b(u, v)$ does not intersect any region in R other than $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$, $p' \in \texttt{Vor}(U)[u] \cup \texttt{Vor}(U)[v]$. This implies there exists burn paths $\pi(u, p')$ and $\pi(v, p')$ in R. □

Observation 4 *Without loss of generality, assume* R *is enclosed by* $\partial P(v, u)$.

In Case 1 of Lemma 12, the existence of p' *and burn paths* $\pi(u, p')$, $\pi(v, p')$, *and* $\pi(v', p')$ *in* R *implies the existence of two new enclosed regions derived from* $\texttt{Vor}(U)$ *(Fig. 9):*

- $A = \mathcal{R}(\{v, v'\}, p', (v, v'))$
- $B = \mathcal{R}(\{u, v'\}, p', (v', u))$

In case 2 of Lemma 12, the existence of p' *and burn paths* $\pi(u, p')$ *and* $\pi(v, p')$ *implies the existence of two new enclosed regions derived from* $\texttt{Vor}(U)$

- $A = \mathcal{R}(\{v\}, p', (v, p'))$
- $B = \mathcal{R}(\{u\}, p', (p', u))$

Observation 5 *The region* $R_{u,v} = R \setminus (A \cup B)$ *is enclosed by burn paths* $\pi(u, p')$, $\pi(u, p)$, $\pi(v, p')$, *and* $\pi(v, p)$. *Furthermore,* $R_{u,v}$ *lies entirely in* $\texttt{Vor}(U)[u] \cup \texttt{Vor}(U)$ *and is isolated from* $A \cup B$ *by* $\pi(u, p') \cup \pi(v, p')$.

Theorem 10 *Given an enclosed region* $R = \mathcal{R}(N, p, (a, b))$, *a time* $t(p_0)$ *and* v' *as the next enclosing vertex, if it is possible, the Next Enclosing Vertex Algorithm gives us two enclosed regions* A *and* B *separated by* $\pi(v', p')$, *such that* A *and* B *being properly burned in time* $t(p_0)$ *by a set of ignition points* U *is equivalent to* R *being properly burned in time* $t(p_0)$ *and paths* $\pi(N, p')$ *and* $\pi(v', p')$ *being burn paths in* $\texttt{Vor}(U)$. *If it is not possible the Next Enclosing Vertex Algorithm returns nothing.*

Proof. In both cases, A and B are separated by $\pi(v', p')$ (Obs. 3 & 4). This implies that if A and B are properly burned, v' must be an ignition point.

If $d(v', p) < d(N, p)$, p would not be in the Voronoi regions associated with enclosing vertices in N which means the enclosing vertices and enclosing point of R would not be preserved if v' is chosen as an ignition point (Step 1). In addition, if v' implies an enclosing point p' that is not burned by v' in time $t(p_0)$, v' cannot be the next enclosing vertex (Step 2a for both cases).

- Case 1: Without loss of generality, assume $R = \mathcal{R}(\{u\}, p, (p, u))$.
  $p' \in \partial P(p, v)$ since if this was not the case, it would imply $d(v', p) < d(u, p)$ since $b(u, v')$ would properly intersect $\pi(u, p)$ which would imply p is closer to v' than u. If A and B are properly burned in time $t(p_0)$, it must be the case that the burn path $\pi(u, p')$ exists. Let R_u be the region enclosed by burn paths $\pi(u, p')$ and $\pi(u, p)$ and $\partial P(p, u)$. By Obs. 1 (star-shaped property), R_u lies entirely in $\texttt{Vor}(U)[u]$. This means $\pi(u, p')$ separates $\pi(u, p)$ from $A \cup B$ which implies that $\pi(u, p)$ is preserved. $\partial P(p, u)$ must be burned by u in time $t(p_0)$ (Step 2b). Since burn paths $\pi(u, p')$ and $\pi(u, p)$ exist and $\partial P(p, p')$ can be burned by u in time $t(p_0)$, by Lemma 1, R_u is entirely burned by u in time $t(p_0)$. This means $R = A \cup B \cup R_u$ is entirely burned in time $t(p_0)$ which implies R is properly burned.
- Case 2: Without loss of generality, assume $R = \mathcal{R}(\{u, v\}, p, (v, u))$.
  $p' \in R$ since if this was not the case, similar to the previous argument, v' would be too close to p. If A and B are properly burned in time $t(p_0)$, it must be the case that burn paths $\pi(u, p')$ and $\pi(v, p')$ exist. This implies that burn paths $\pi(u, p)$ and $\pi(v, p)$ are preserved since they are isolated from A and B by $\pi(u, p') \cup \pi(v, p')$ (Obs. 5. Since p and p' are burned by u and v in time $t(p_0)$, the segment S of $b(u, v)$ between p and p' is burned by u and v by Lemma 1. Let $R_{u,v}$ be the region enclosed by burn paths $\pi(u, p')$, $\pi(u, p)$, $\pi(v, p')$, and $\pi(v, p)$. By Obs. 5, $R \setminus (A \cup B) \subset R_{u,v}$. Similar to Case 1, since S is burned by u and v, the region enclosed by burn paths $\pi(u, p)$ and $\pi(u, p')$ and S is entirely burned by u (and similar for v) (Lemma 1). This implies $R_{u,v}$ is entirely burned by u and v in time $t(p_0)$. This means $R = A \cup B \cup R_{u,v}$ is entirely burned in time $t(p_0)$ which implies R is properly burned.

In both cases, if either A or B cannot be properly burned in time $t(p_0)$, since each region is isolated by known burn paths, R cannot be entirely burned in time $t(p_0)$ with v' as the next enclosing vertex. □

E Determining if an enclosed region is entirely burned by a set of enclosing vertices in time $t(p_0)$

This method takes the arguments to an enclosed region $R = \mathcal{R}(\{N\}, p, (a, b))$ and a time t to burn the enclosed region with N. We assume that p is burned by N in time t.

Implementation:

There are two cases depending on $|N|$:

- $|N| = 1$**:** If $d(u, v') \leq t$ for all $v' \in (V \cap \partial P(a, b)) \cup \{p\}$, return $true$; else return $false$.
- $|N| = 2$**:** It must be the case that $N = \{a, b\}$. Assume, without loss of generality that $u = a$ and $v = b$.

  Let p' be the Type II potential final burn point associated with u and v (clockwise from u to v).

  If $d(u, u') \leq t$ for all $u' \in (V \cap \partial P(u, p')) \cup \{p, p'\}$ and $d(v, v') \leq t$ for all $v' \in V \cap \partial P(p', v)$, return $true$; else return $false$

In both cases, we check all potential points p as in Lemma 3. If all of these points are within t of N, we know that R is entirely covered or we would violate the lemma. If one of the points isn't covered, then R is not covered by N.